

VidaReal Translator

Automatic AI translation of the Rock RMS interface with permanent caching — built in-house, without modifying Rock's core.

Executive brief · August 2026 · Plugin version 1.2.0

1 Executive summary

Rock RMS — the platform that manages the people, groups, events and check-in of the church — ships **in English only**. That slows adoption among staff and volunteers. **VidaReal Translator** translates the entire interface into Spanish (or any language) automatically and transparently for the user.

The core idea: every interface string is translated by AI **once** and is then stored in our own database. From the second user onward, the translation comes out of the cache — instant and free. AI spend trends to zero as usage grows.

Once

each string's translation is paid for a single time; after that it is permanently cached

Zero changes

to Rock's code: a standalone plugin, updatable and reversible

Interface only

sending member data to the AI is blocked by design

One click

to switch it on, or remove it from every Rock site at once

2 What the user sees

- **The whole interface in Spanish:** menus, buttons, forms, validation messages and help text — on the intranet, the public site and check-in.
- **Optional language switcher** (Weglot-style): a discreet tab to switch between Spanish, English or other languages. The preference is remembered per device.
- **Zero friction:** nothing to install, no change to their workflow. If anything fails, the page simply shows in English — it never breaks.

3 Architecture

Four pieces, each with one clear responsibility. The numbered arrows follow the journey of a new string:

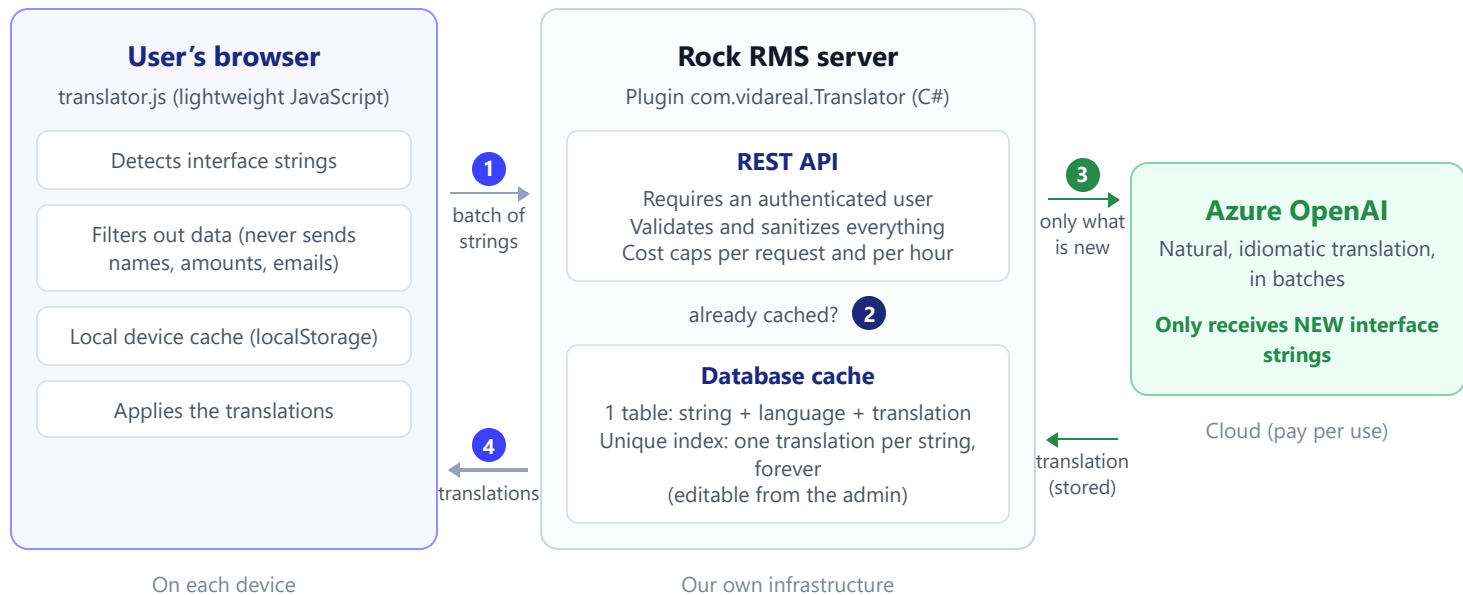


Fig. 1 — Architecture. The browser never talks to Azure directly; everything goes through the Rock server, which validates, applies caps and caches.

Components

Component	Responsibility
translator.js (browser)	Detects the visible interface strings live — including classic screens (WebForms), modern ones (Obsidian/Vue) and AJAX-loaded content — and applies the translations without reloading. Includes the language switcher.
REST API (C# plugin)	Receives batches of strings, resolves them against the database cache and only sends the AI what has never been translated. Requires an authenticated user; sanitizes the AI output; enforces cost limits.
Database	A single translations table with a unique index on (string, language). It is the permanent memory of the system and is managed from within Rock.
Azure OpenAI	Translates software vocabulary (buttons, menus, messages) with human-level quality, under strict instructions: never touch proper names, codes or variables.

4 How it works at the DOM level

The translator touches no server-side HTML or templates: it operates on the **already-rendered DOM** in the browser, as a visual layer. Every string goes through a 5-step pipeline:

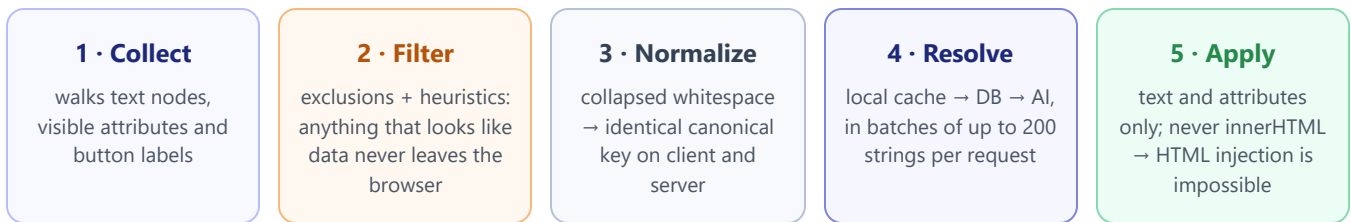


Fig. 2 — Pipeline in the browser. Steps 1–3 and 5 are local; only step 4 can touch the network.

Anatomy of a page: what it touches and what it does not

```
<label>First Name</label> ✓ translates interface text
<input value="Juan Pérez"> × untouched form value: whatever is typed or saved is never touched
<input type="submit" value="Save"> ✓ translates here value is the visible button label, not data
<div title="Print this page"> ✓ translates visible attributes: title, placeholder, aria-label, alt
<table class="grid-table">
  <th>Email Address</th> ✓ translates column header = interface
  <td>juan@vidareal.tv</td> × untouched data cell: excluded wholesale, never even sent
<span>{{ Person.NickName }}</span> × untouched Lava token / variable
<textarea>...</textarea>, <pre>, <code>, contenteditable × untouched editable or technical content,
excluded wholesale
```

In addition, any element can be excluded individually with the class **nottranslate** or the attribute **data-no-translate**, and there are selector lists configurable from the admin without recompiling.

- **Idempotent and efficient:** nodes that have already been translated are marked and never reprocessed; strings already known come out of the cache without touching the network.
- **Safe application:** text is replaced node by node, preserving edge whitespace; innerHTML is never used, so the AI cannot introduce HTML or scripts.
- **Fail-safe:** every step runs inside error guards; an invalid selector or an odd response is discarded without affecting the page.

Coverage of dynamic content

Rock builds its pages from blocks that arrive at different moments. The translator covers all four paths:

Scenario	Mechanism
Initial load and late render	Staggered passes (0 / 600 / 1500 ms) over the whole page.
Content injected by AJAX / modern blocks (Obsidian)	MutationObserver : watches the DOM and processes <i>only the new subtrees</i> (it does not re-sweep the page), grouping changes with a small debounce.
Partial postbacks (classic WebForms screens)	Hook into the ASP.NET AJAX lifecycle (PageRequestManager): re-scans when each postback finishes.
Navigation without reload (SPA)	Hook into the History API (pushState / popstate): re-scans on route change.

5 The cache funnel: why the cost trends to zero

Every string on the page passes through three levels. Only the last one costs money, and each string crosses it **once in the lifetime of the system**:

100% of the strings on the page

every visit



With use, levels 1 and 2 absorb practically everything: the Rock interface is finite, and once it is translated, level 3 is barely visited.

Fig. 3 — Resolution funnel. Cost lives only at level 3, and each string pays it exactly once.

Built-in cost controls

Control	Effect
Permanent cache (DB + browser)	Each string is paid for once in the lifetime of the system.
Per-request and hourly limits	A maximum number of strings per request and a global cap on new translations per hour: an uncontrolled spending spike is impossible.
Deduplication and anti-repetition	Strings repeated on the page and simultaneous requests are consolidated: the same thing is never paid for twice.
No credentials → no spend	If Azure is not configured, the plugin simply does not translate. It never generates accidental spend.
Authenticated access	Only users with a Rock session can invoke the service: no outsider can consume the budget.

6 Protection of member data

A non-negotiable design rule: «**interface yes, data no**». The filter runs in the browser *before* anything is sent: whatever looks like data never leaves the device.

✓ Translated	× Never sent, never touched
Buttons, menus, titles, form labels, validation messages, help text	People's names, emails, phone numbers, amounts, dates, the contents of data tables, fields that are typed in or saved, editable content

- The values that travel to the server are excluded by design: a translation **cannot corrupt** what gets stored in the database.
- The AI response is sanitized (any code or markup is stripped) before being stored or displayed.
- Aligned with the Responsible AI Use Policy: only software vocabulary ever reaches the AI, never member information.

7 Administration (inside Rock itself)

Everything is managed from a single admin page: *Admin Tools* → *Installed Plugins* → *VidaReal Translator*. No need to touch servers or databases.

Function	Description
On / off	A single toggle activates the translator on every Rock site at once ; switching it off removes it just as cleanly. Fully reversible.
Editorial control	A listing lets you search any translation and correct it by hand (the correction is locked in: the AI will not touch it again), exclude strings that must never be translated, or delete them to regenerate.
Cache purge	Regenerates every translation (e.g. after improving the AI style instructions). Administrators only.
Configuration	Target language, Azure credentials (stored encrypted), the languages in the switcher and fine-grained scope settings — all through the standard Rock form.

8 Quality and robustness

- **Graceful degradation:** in the face of any failure (AI down, network, configuration), the page is shown in English. The translator never breaks a screen or a save.
- **Professional translation:** the AI is instructed to use standard software terminology in Spanish, a consistent tone and a similar length (so the layout does not break).
- **Verified integrity:** automated tests cover the data safeguards and the exact parity between the browser code and the server code.
- **Built to grow:** it supports several languages in parallel (each with its own cache) and the architecture allows switching AI provider without rewriting the system.

In summary

A one-time development investment that makes Rock accessible in Spanish for the whole team, with a marginal and decreasing operating cost (trending to zero), full editorial control and zero risk to the data of the congregation.

The same architecture works for any target language: each one keeps its own permanent cache, and adding one does not change a single line of Rock.